IFSC 3360: System Analysis and Design

Airport Baggage System

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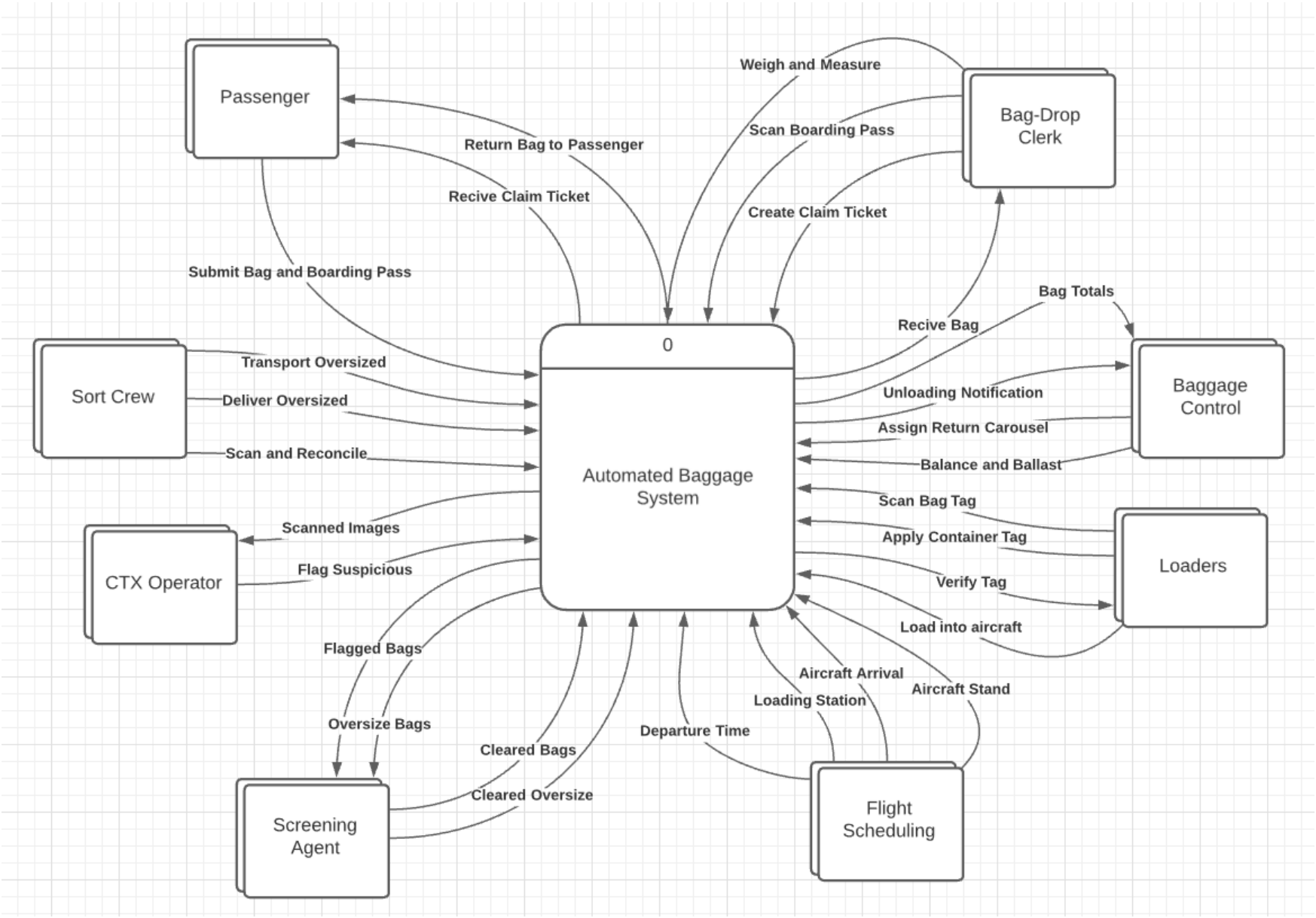
System Description

- An automated baggage system transports luggage to, from and between aircraft
- The information system will provide fast, reliable and auditable routing and tracking for all bags in the baggage system
- This is a new system.
  - I'd rather start from scratch
  - Systems like this need to be customized for each installation anyway.

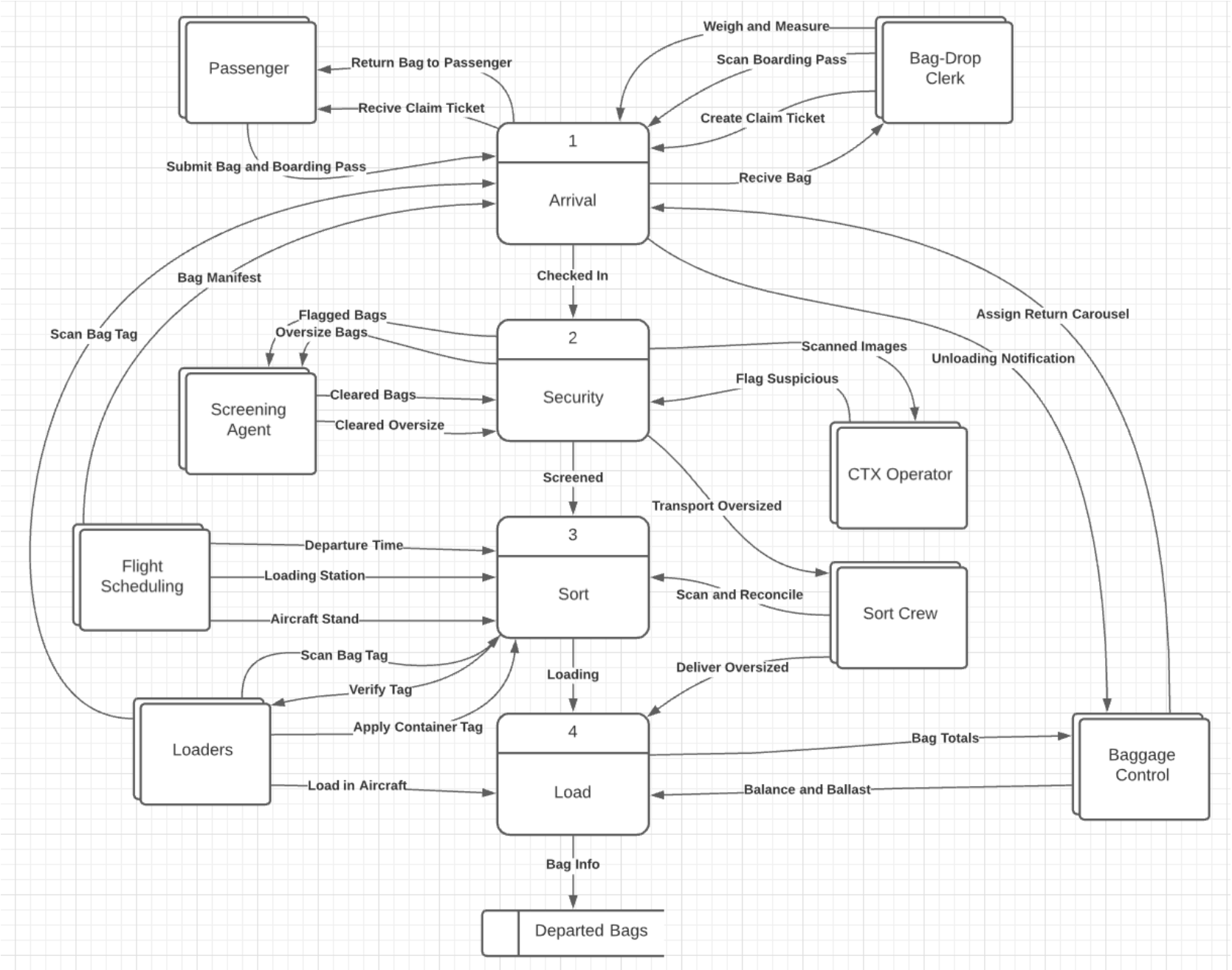
System Requirements

- The system shall do
  - Continuously track each bag in the physical system
  - Update the status of each bag as it proceeds through the system
  - Ensure delivery of each bag to the correct destination in a timely manner
  - Accommodate rush and oversize luggage
  - Facilitate proper aircraft loading
  - Respect airport security standards

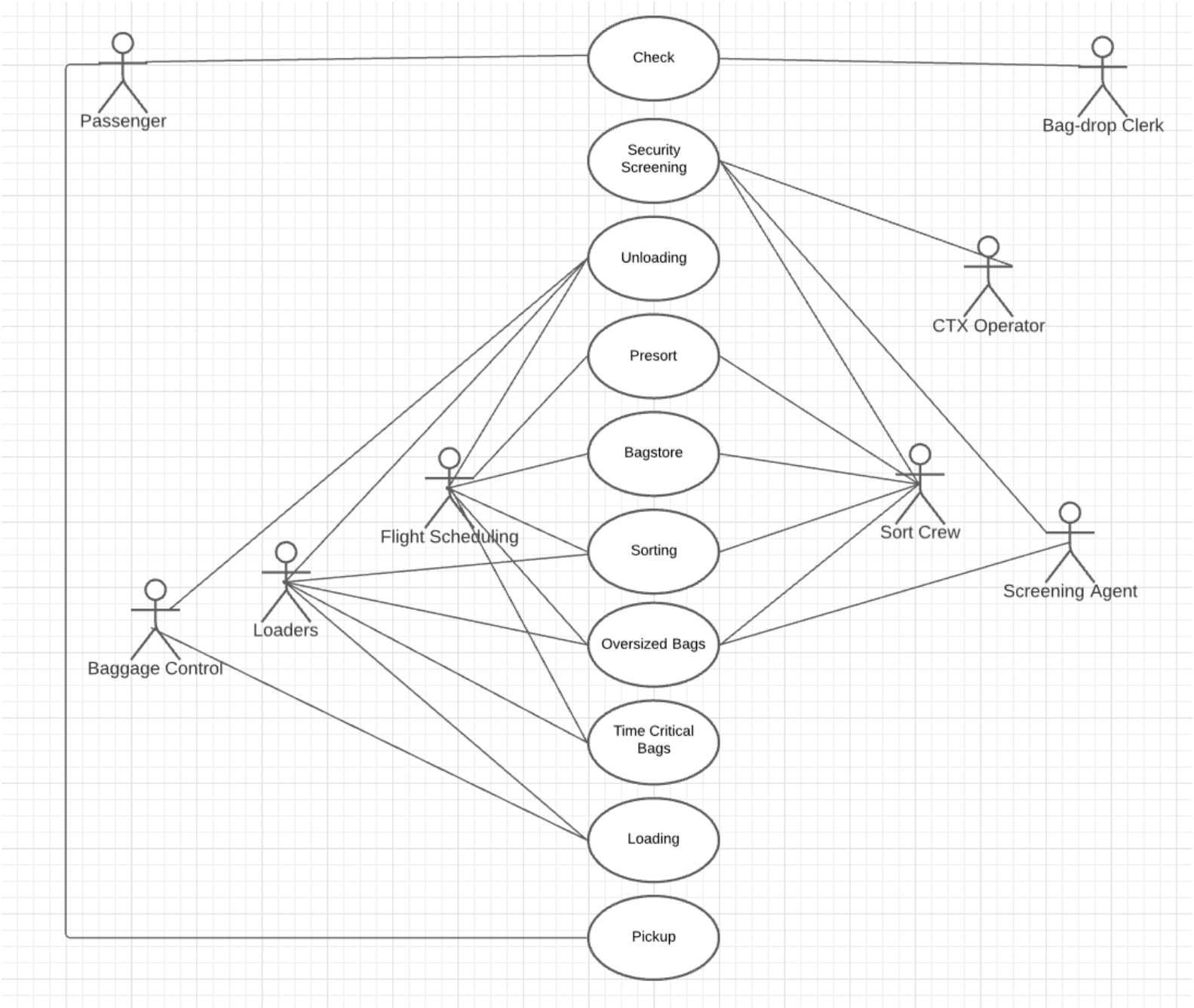
Context Diagram



DFD 0 Diagram



Use Case Model



Use Case Model

UC1: Check In
Actors: Passenger, Bag-drop clerk

Description: The use case submits a bag into the system

Main success scenario:

1. Clerk greets passenger
2. Passenger submits boarding pass
3. Clerk scans boarding pass
4. Passenger submits bag
5. Clerk weighs and measures bag
6. System prints bag ticket and claim ticket
7. Bag status initialized as “checked-in”
8. Clerk applies bag ticket to bag
9. Clerk submits bags into automated system
10. Clerk gives claim ticket to passenger

Alternate Scenarios:

3a. Boarding pass will not scan
    1. Hand key pass number

3b. Pass number not recognized
    1. Refer passenger to customer service

5a. Bag overweight
    1. Additional charge to passenger

5b. Bag oversize
    1. Additional charge to passenger
    2. Route bag directly to aircraft stand

5c. Customer refuses additional charges
    1. Bag cannot be checked
    2. Call supervisor for support if needed

6a. Tickets fail to print
    1. Refill printer
    2. Call help desk

Preconditions
    1. Passenger has valid boarding pass

Postconditions
    1. Passenger receives claim ticket from clerk
    2. Passenger continues on to flight
    3. Bags proceed to UC2 security screening